



University of Missouri

The Missouri Growth Model – An Overview

Brief Growth Model History

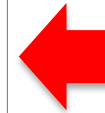
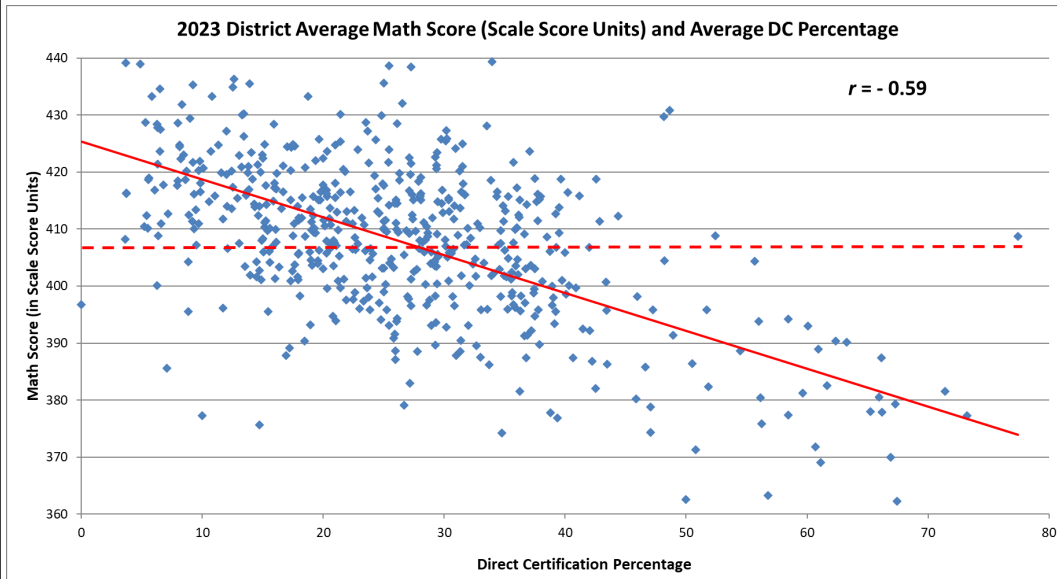
- By mandating grade-level examinations and supporting longitudinal data systems, NCLB created the data environment that allowed for the proliferation of modern growth measures for use in policy and research.
- Forty-eight states and the District of Columbia are measuring and reporting some form of student growth under ESSA (DCQ, 2021).
- DESE convened a Growth Model Technical Advisory Committee in 2011 and 2012 that included many well-respected scholars in the field – Cory Koedel, Dan McCaffrey, Steve Rivkin, Damien Betebenner, and others.
- As a result, the development of the Missouri Growth Model was a very deliberate and highly-resourced process that resulted in a high quality and well-specified statistical model.

Growth Model Overview

- What the **Missouri Growth Model** **is not**:
 - The **Missouri Growth Model** **does not produce** a **student-level measure** designed to track growth to proficiency or a similar target.
- What the **Missouri Growth Model** **is**:
 - The **Missouri Growth Model** **does produce** an **instructional-unit level measure** (e.g., for a building or LEA) that captures the amount of student learning that can be uniquely attributed to the instructional unit.
 - One can think of it as a **measure of instructional effectiveness**.

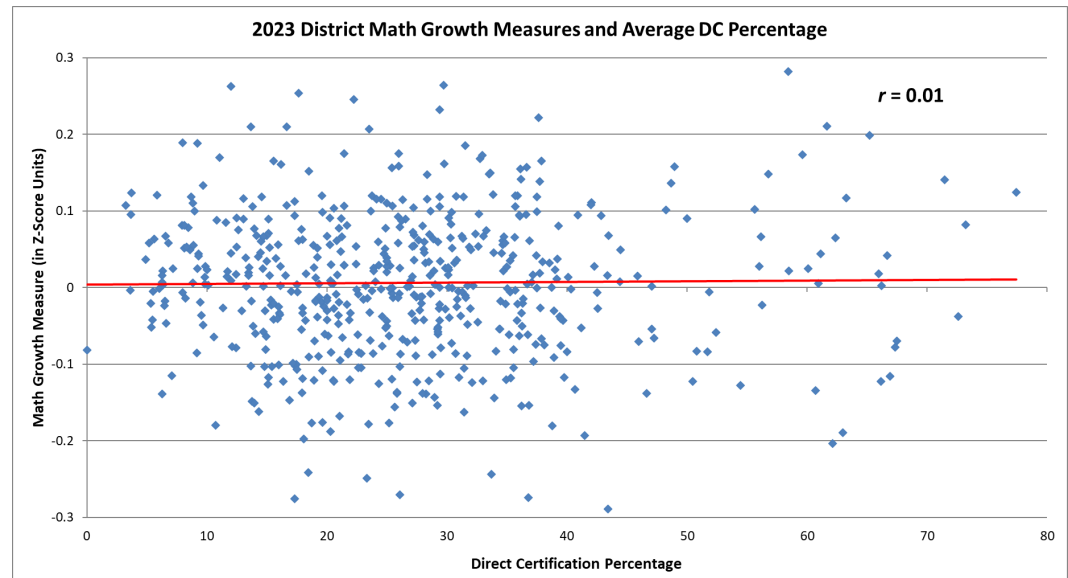
Growth Model Overview

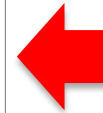
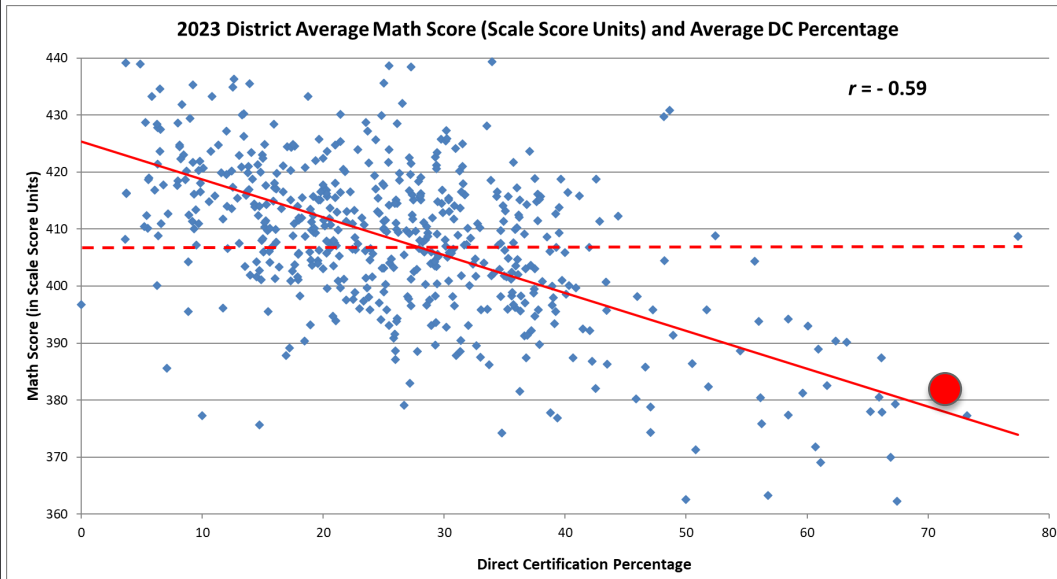
- The **Missouri Growth Model** **does produce** an **instructional-unit level measure** that captures the amount of student learning that can be uniquely attributed to the instructional unit.
- There is a strong research literature providing evidence on this point, including evidence from randomized student assignments to schools and teachers.
 - Kane and Staiger (2008, National Bureau of Economic Research)
 - Deming (2014, American Economic Review)
 - Chetty, Friedman, and Rockoff (2014, American Economic Review)
- In addition to being an accurate measure of instructional effectiveness, growth measures are also more equitable than alternative measures of building and LEA performance.



Average test scores are highly correlated with student poverty. Research shows average test scores capture more about the neighborhood the school serves than the quality of the school itself.

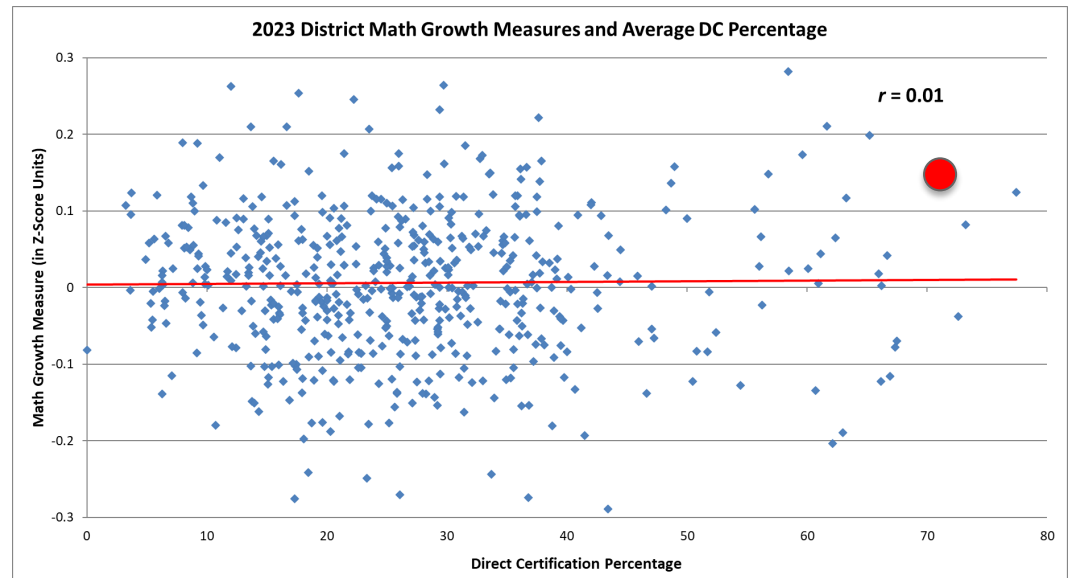
A well-specified growth model breaks (or largely breaks) this correlation. As a result, research shows growth is more informative about instructional quality. Growth measures are also more equitable.



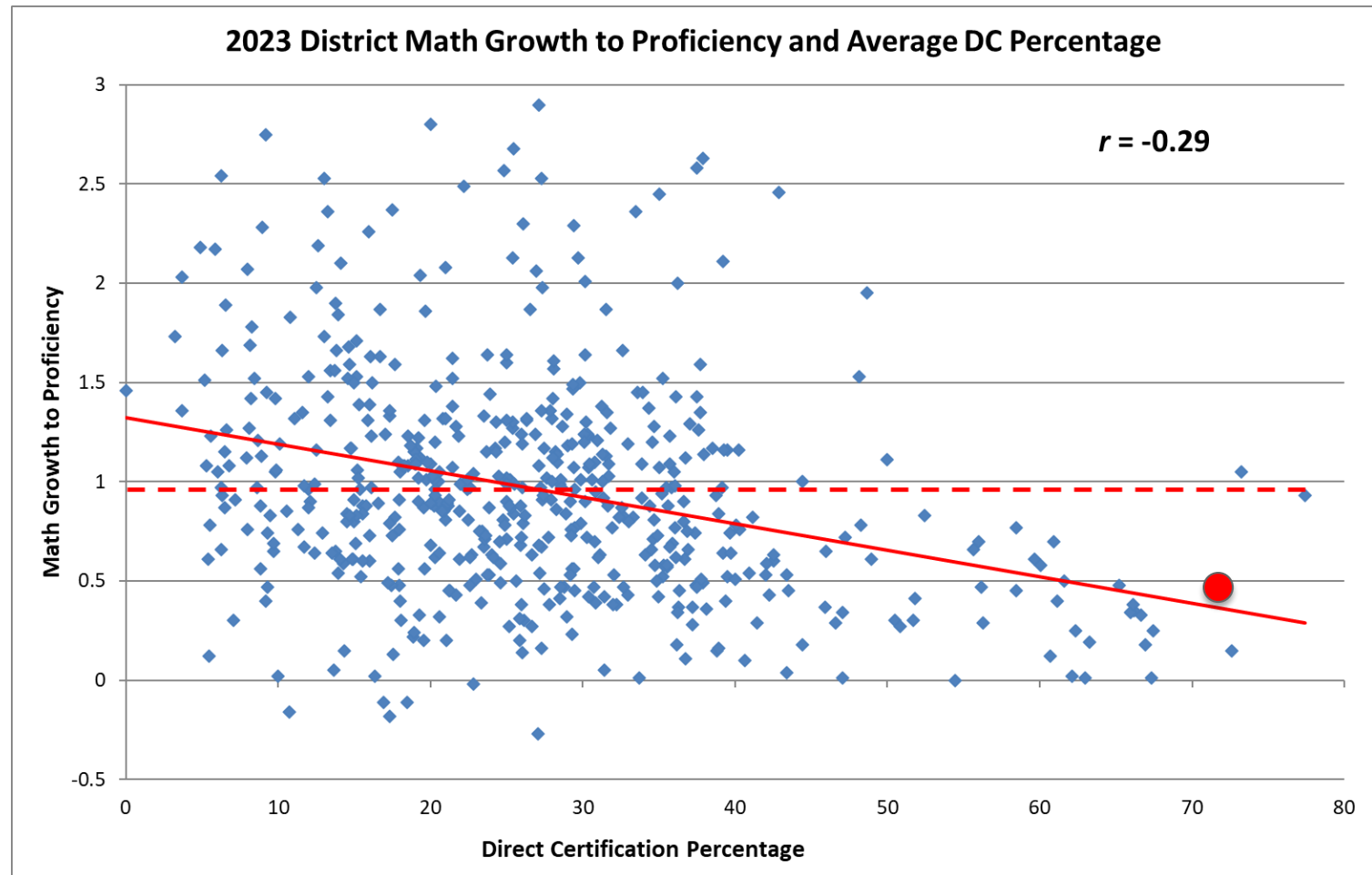


Average test scores are highly correlated with student poverty. Research shows average test scores capture more about the neighborhood the school serves than the quality of the school itself.

A well-specified growth model breaks (or largely breaks) this correlation. As a result, research shows growth is more informative about instructional quality. Growth measures are also more equitable.



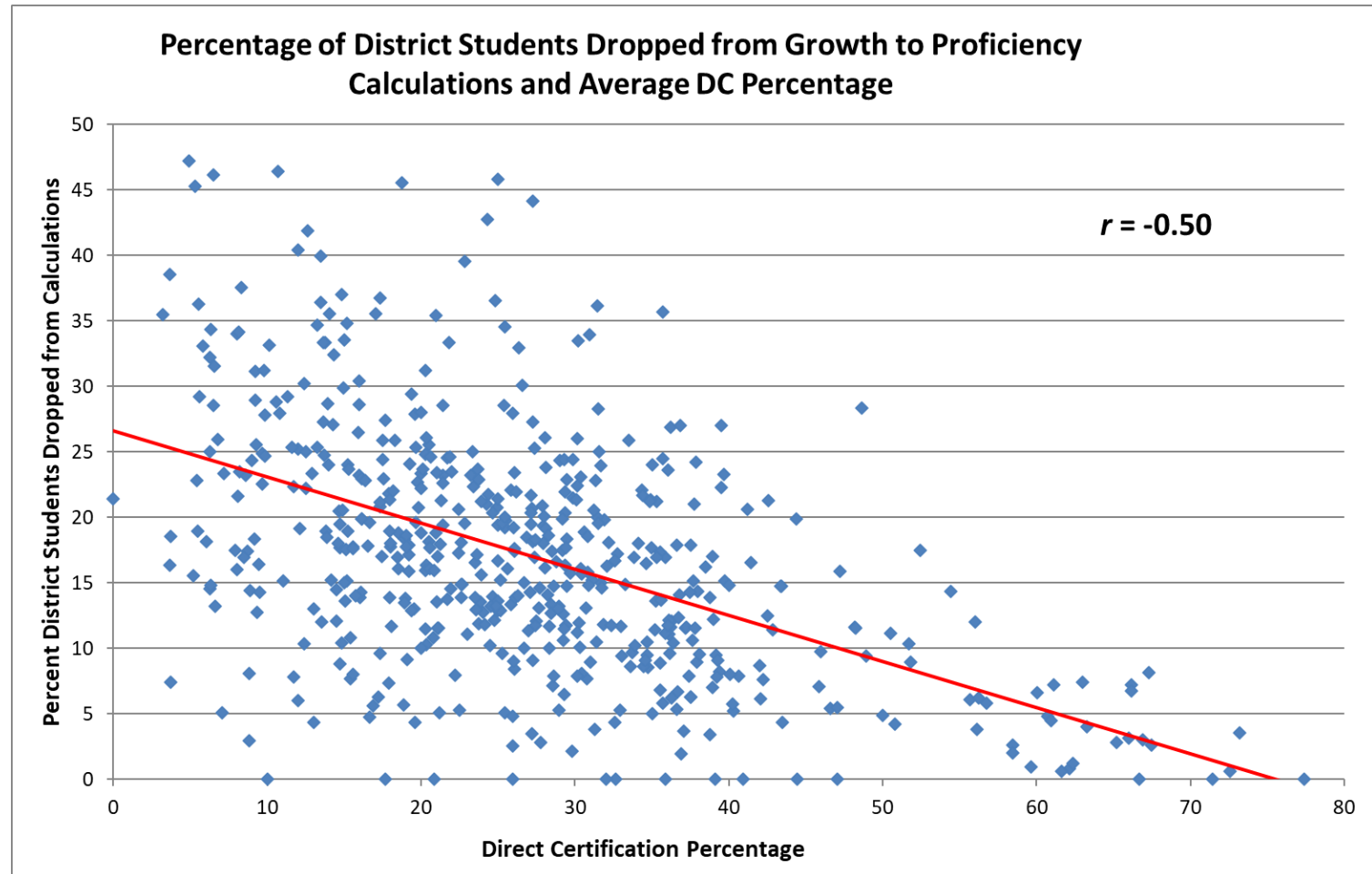
What About Growth to Proficiency?



Other Issues with Growth to Proficiency

- Determining targets is not necessarily straightforward
- Reintroduces “Bubble Students”, as students who are already close to the target scale score the previous year can produce very large growth to proficiency values
- What do you do with:
 - Students who gain more than their gains targets?
 - Students with negative gain scores?
 - And most problematically, students with negative gains targets? If you choose to drop them, you get . . .

What About Growth to Proficiency?



Two Baseball Analogies

- Growth : Status :: Wins Above Replacement : RBIs
- Can monitoring your batting average teach you how to improve your hitting?
 - The number is an indicator, not an educator.

How Does the Growth Model Isolate the School's Unique Contribution

- Consider a simple way to measure student learning:
$$\text{Measured Learning} = \text{Test Score This Year} - \text{Test Score Last Year}$$
- This measure consists of two parts:
 - True Student Learning
 - Noise
- We will return to the noise component later.

How Does the Growth Model Isolate the School's Unique Contribution

- The true learning component also can be broken down into two parts:
 - Factors outside the school's control, e.g.,:
 - The inherent ability of the student
 - Parental resources
 - Neighborhood effects
 - Factors within the school's control, e.g.,:
 - Teacher quality
 - Curriculum choices
 - Administrative policies
- Research shows that the independent variables included in a well-specified growth model are sufficient to control for these factors . . .
- Which means that what is left over can be attributed to the school. This is the growth measure.

Missouri Growth Model Calculation

- In the first stage of the Missouri Growth Model estimation, a statistical model uses all the exam scores in the state to determine the *statewide relationship* between current year exam scores and previous year exam scores.

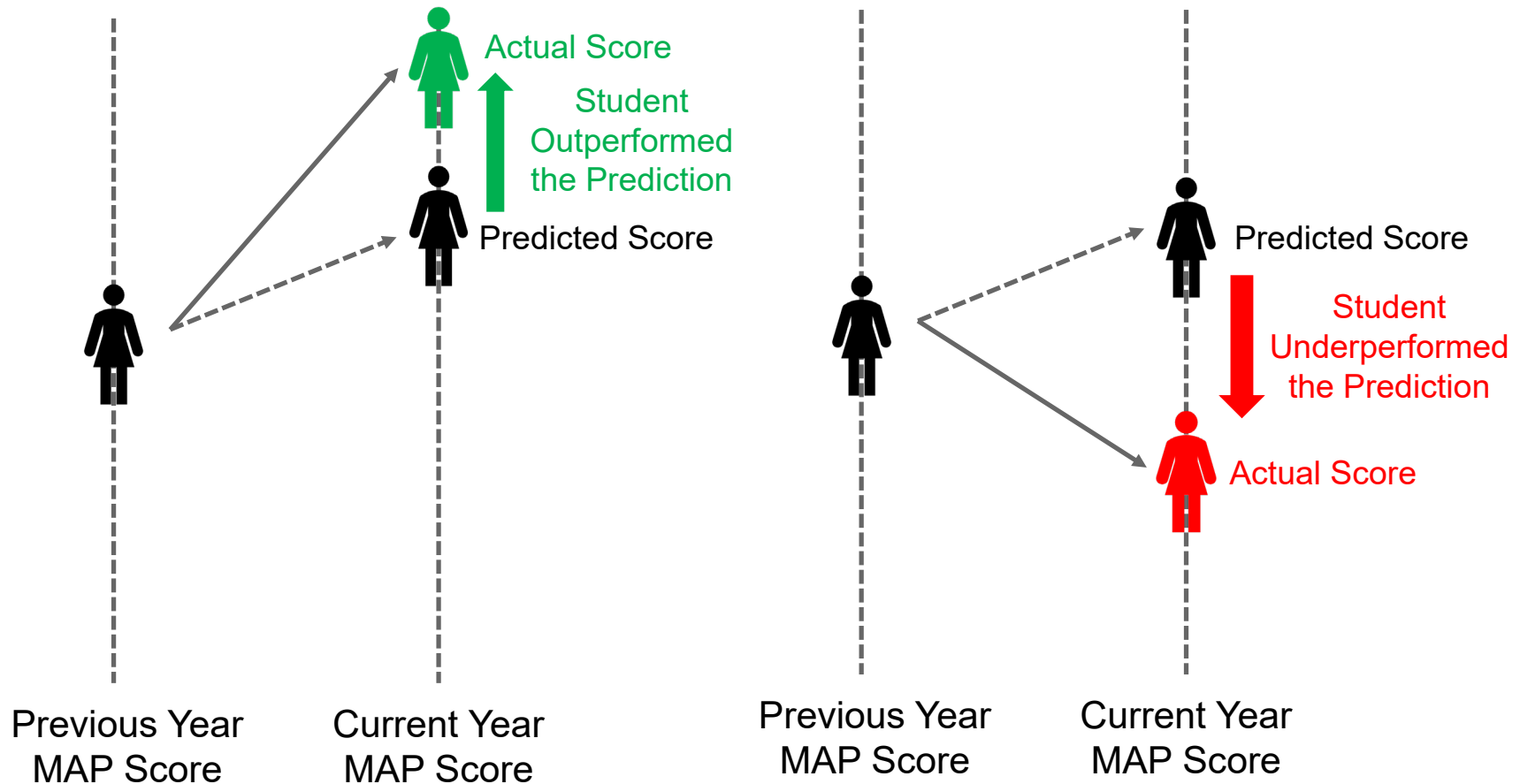
Missouri Growth Model Calculation

- After the first-stage model is estimated, each student has a predicted score in each subject based on the statewide relationship between previous and current exam scores.
- This predicted score is then compared to a student's actual score, and a student-level growth measure (student residual) for each student is calculated as:

$$(\text{Student Residual} = \text{Observed score} - \text{Predicted score})$$

- As a result,
 - If the student residual is positive, the student performed **better than predicted** on the exam based on statewide patterns.
 - If the student residual is negative, the student performed **worse than predicted** on the exam based on statewide patterns.

Missouri Growth Model Calculation



Missouri Growth Model Calculation

- The building and LEA growth measures are calculated directly from the student residuals.
- They are simply the average of the residuals for the students who took the MAP test in each building/LEA in the current year.
- Hence,
 - A **positive** growth measure indicates that, on average, students in the building/LEA performed **better** than their predicted scores.
 - A **negative** growth measure indicates that, on average, students in the building/LEA performed **worse** than their predicted scores.

Missouri Growth Model Calculation

- For each growth measure, a statistical significance category is also reported. This is based on:
 - How far the estimate is away from zero (or average) – The further away from zero, the more likely it is statistically significant.
 - Consider flipping a coin and trying to determine whether it is fair. The further the percentage of heads gets away from 50%, the more confident you are that the coin is not fair.

Missouri Growth Model Calculation

- For each growth measure, a statistical significance category is also reported. This is based on:
 - How far the estimate is away from zero – The further away from zero, the more likely it is statistically significant.
 - How much noise is in the measure, which is largely driven by the number of student records included in its estimation – The noisier the estimate, the less likely it is statistically significant.
 - In the coin flipping example, the more times you flip the coin, the more confident you become about whether it is fair or not.
 - If you only flip the coin four times, and it comes up heads 3 out of 4 times, you cannot be confident in its fairness.
 - In contrast, if you flip the coin 1000 times and it comes up heads 750 times, then you can be confident the coin is not fair.

Missouri Growth Model Calculation

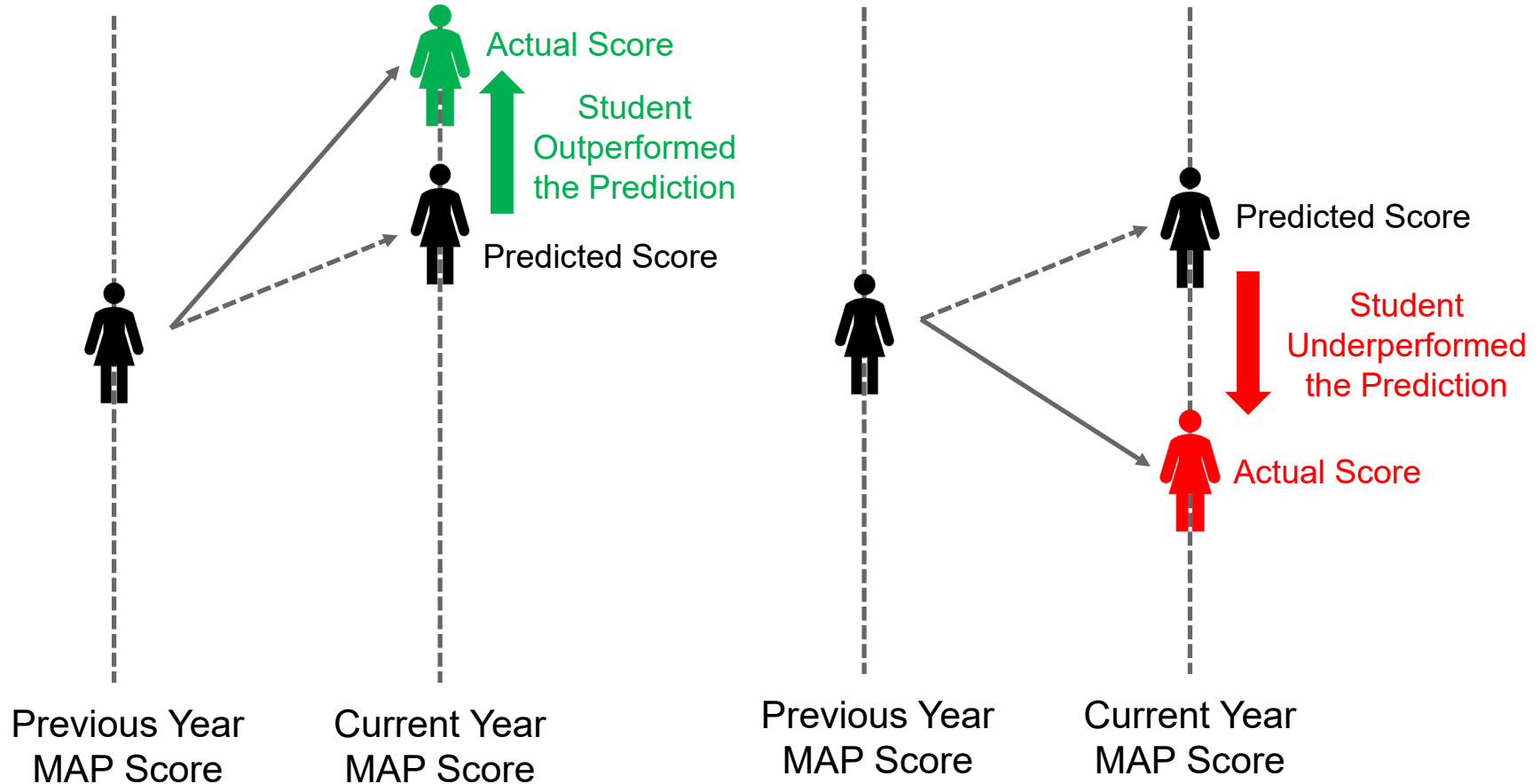
- For each growth measure, a statistical significance category is also reported.
- It is worth noting, however, that all the MSIP metrics (particularly those derived from test scores) contain statistical noise. However, growth is the only one for which statistical significance is reported/considered in MSIP calculations.
- So, the use of statistical significance in MSIP scoring formulas is, in some ways, holding growth to a higher statistical standard than the other metrics.

Missouri Growth Model Calculation

- First-Stage Technical Notes:
 - The statewide predictive model is estimated separately by grade and subject.
 - Both previous year subjects are included as predictors in every model.
 - Statistical adjustments are made to account for students with a missing exam score in one of the two previous-year subjects.
 - Previous year exam scores are also averaged and included at the building and LEA levels.
 - Other explanatory variables used in the first-stage model include flags for student mobility and neglected/delinquent status.

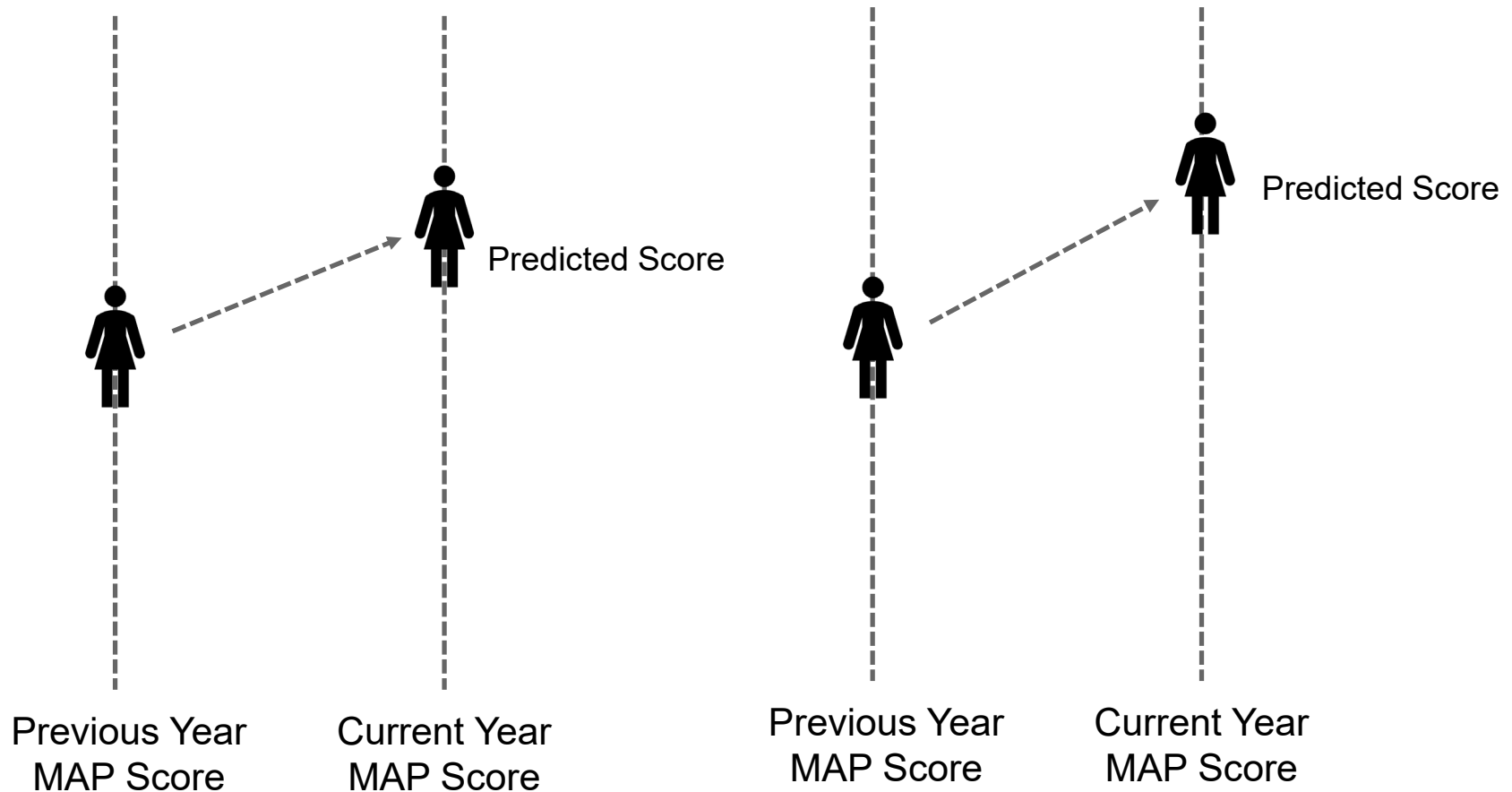
Missouri Growth Model Overview

Two Students – Same Prior Score(s), Similar School Environment



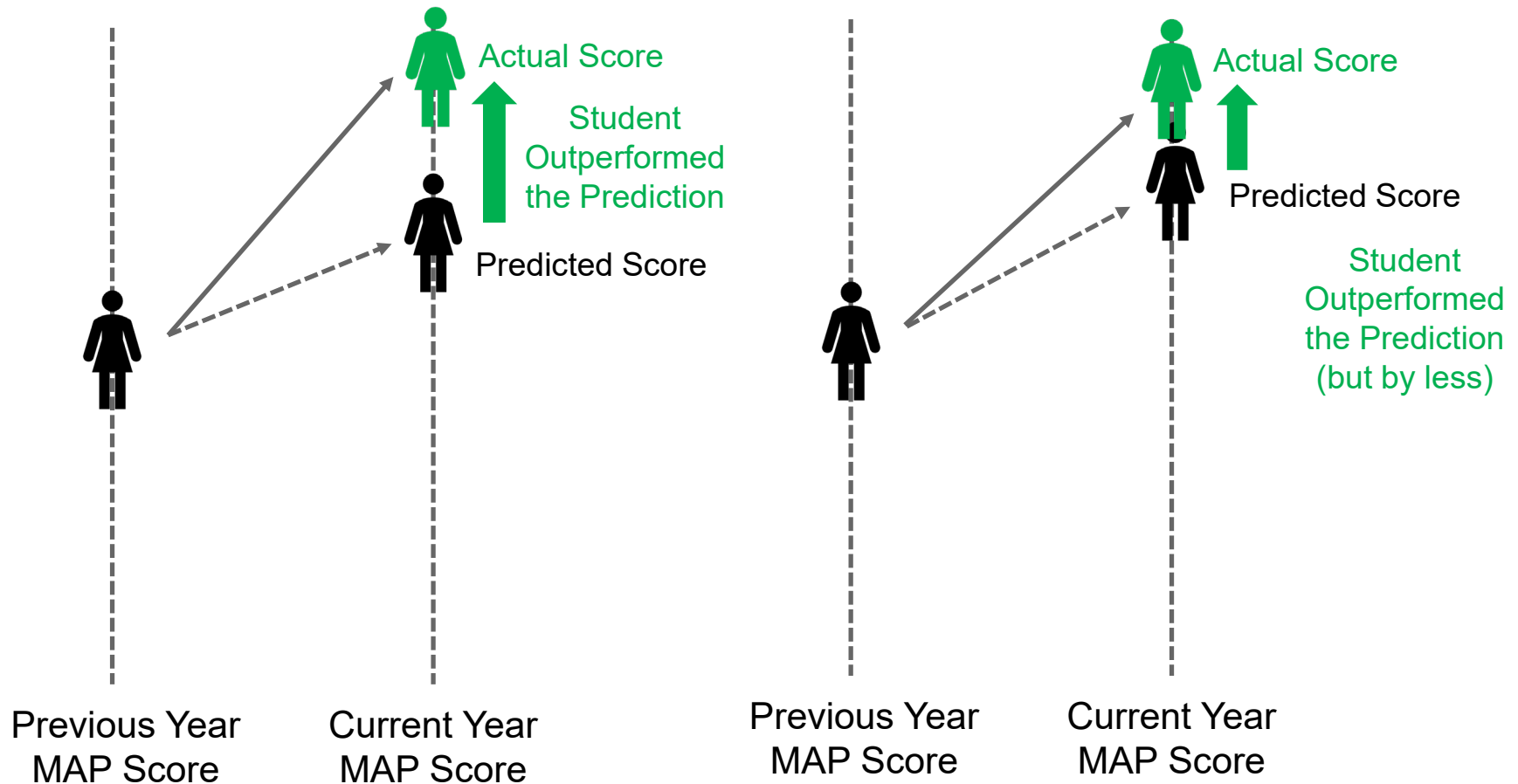
Missouri Growth Model Overview

Two Students – Same Prior Score(s), Different School Environment



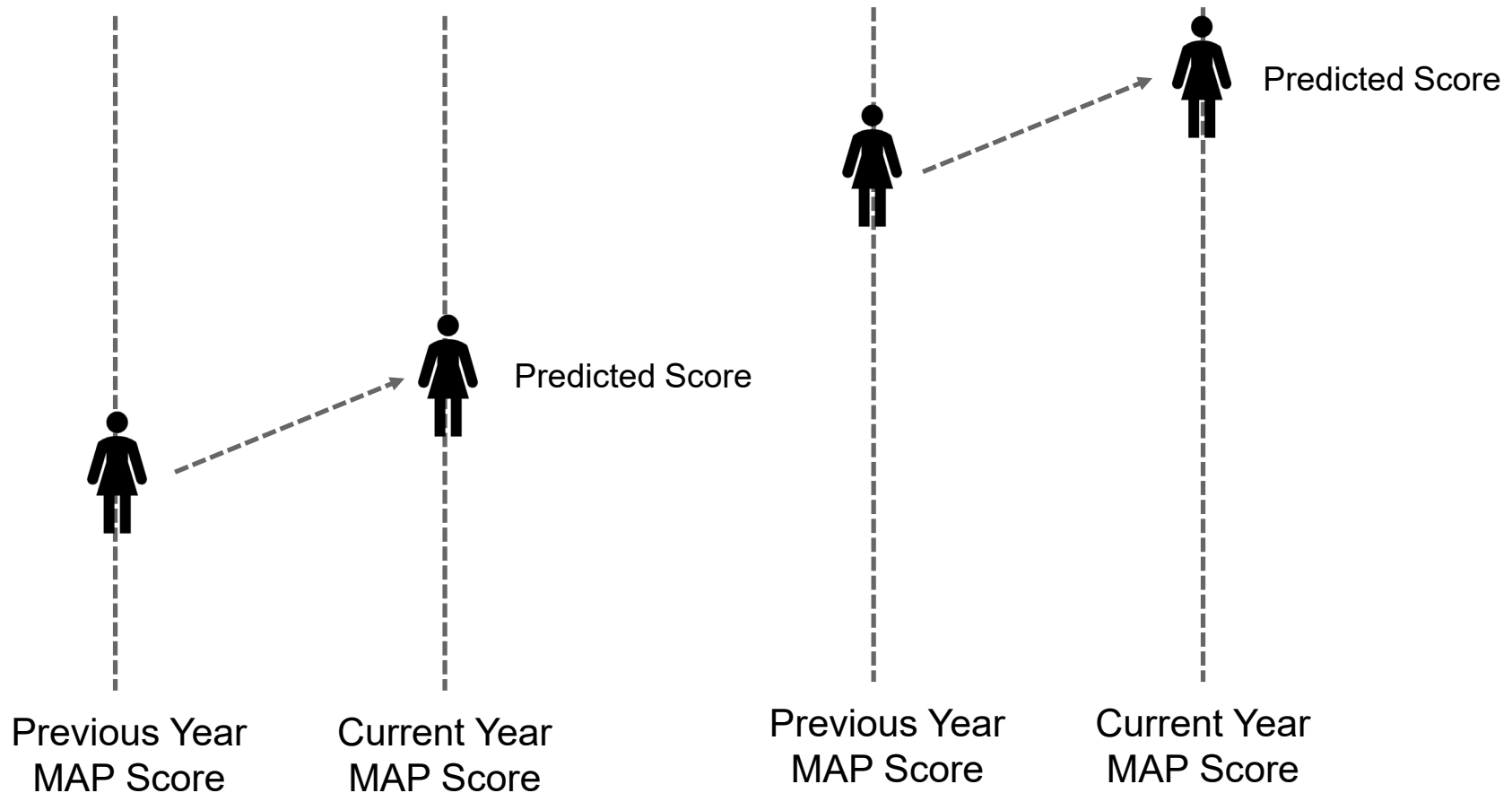
Missouri Growth Model Overview

Two Students – Same Prior Score(s), Different School Environment



Missouri Growth Model Overview

Two Students – Different Prior Score(s), Similar School Environment



Missouri Growth Model Calculation

- Second-Stage Technical Notes:
 - After the student residuals are averaged to produce the raw building or LEA growth measures, a further statistical adjustment (Bayesian shrinkage) is made to account for the noise inherent in the measures.
 - Subgroup growth measures are also re-centered on the overall state average student residual for the subgroup, so that the average of the subgroup growth measures is (near) zero.

Missouri Growth Model – Residuals File

- Second-Stage Technical Notes:
 - After the student residuals are averaged to produce the raw building or LEA growth measures, a further statistical adjustment (Bayesian shrinkage) is made to account for the noise inherent in the measures.
 - Subgroup growth measures are also re-centered on the overall state average student residual for the subgroup, so that the average of the subgroup growth measures is (near) zero.